B. Tech. – M. Tech. Computer Science and Engineering (Cyber Security)

5-Year Integrated Program

Teaching & Examination Scheme

Semester I													
Sr. No.	Subject Code	Subject Name	Teaching Scheme (Credit)					Examination Scheme						
								Hours			Component Weightage			
			L	T	P	C	TCH	MSE	SEE	PRE	MSE	TA	LPW	SEE
1	CTBTCSE SI P1	Engineering Physics	2	0	0	2	2	01:30	03:00	-----	0.2	0.2	-----	0.6
2	CTBTCSE SI P2	Engineering Mathematics - 1	3	1	0	4	4	01:30	03:00	-----	0.2	0.2	-----	0.6
3	CTBTCSE SI P3	Basics of Electrical Engineering	2	1	0	3	3	01:30	03:00	-----	0.2	0.2	-----	0.6
4	CTBTCSE SI P4	Fundamentals in Computer Programming with C	3	0	0	3	3	01:30	03:00	-----	0.2	0.2	-----	0.6
5	CTBTCSE SI P5	Engineering Graphics	2	0	0	2	2	01:30	03:00	-----	0.2	0.2	-----	0.6
6	CTBTCSE SI P6	Communication Skills	2	1	0	3	3	01:30	03:00	-----	0.2	0.2	-----	0.6
7	CTBTCSE SI L1	Engineering Physics Laboratory	0	0	1	1	2	-----	-----	03:00	-----	-----	1.0	-----
8	CTBTCSE SI L2	Fundamentals in Computer Programming Laboratory	0	0	1	1	2	-----	-----	03:00	-----	-----	1.0	-----
9	CTBTCSE SI L3	Engineering Graphics Laboratory	0	0	1	1	2	-----	-----	03:00	-----	-----	1.0	-----
		Total Credit & Total Credit Hours				20	23							
Semester II														
Sr. No.	Subject Code	Subject Name	L	T	P	C	TCH	MSE	SEE	PRE	MSE	TA	LPW	SEE
1	CTBTCSE SII P1	Engineering Chemistry	2	0	0	2	2	01:30	03:00	-----	0.2	0.2	-----	0.6
2	CTBTCSE SII P2	Engineering Mathematics – 2	3	1	0	4	4	01:30	03:00	-----	0.2	0.2	-----	0.6
3	CTBTCSE SII P3	Object Oriented Programming with C++	3	0	0	3	3	01:30	03:00	-----	0.2	0.2	-----	0.6
4	CTBTCSE SII P4	Digital Logic Design	2	0	0	2	2	01:30	03:00	-----	0.2	0.2	-----	0.6
5	CTBTCSE SII P5	Professional Ethics	2	1	0	3	3	01:30	03:00	-----	0.2	0.2	-----	0.6
6	CTBTCSE SII P6	Fundamentals of Forensic Science & Laws	2	1	0	3	3	01:30	03:00	-----	0.2	0.2	-----	0.6
7	CTBTCSE SII P7	Environment Science	0	0	0	0	0	-----	-----	-----	-----	-----	-----	-----
8	CTBTCSE SII L1	Engineering Chemistry Laboratory	0	0	1	1	2	-----	-----	03:00	-----	-----	1.0	-----
9	CTBTCSE SII L2	Object Oriented Programming with C++ Laboratory	0	0	1	1	2	-----	-----	03:00	-----	-----	1.0	-----

SEMESTER – I

CTBTCSE SI P1: Engineering Physics

Teaching Scheme					Evaluation Scheme								
					Theory						Practical		
					Internal Exams				University Exams		University Exams (LPW)		Total
					TA-1/TA-2		MSE						
Th	Tu	Pr	C	TCH	Marks	Hrs	Marks	Hrs	Marks	Hrs	Marks	Hrs	
2	0	0	2	2	25	00:45	50	01:30	100	03:00	-	-	200

* Note: TA-2 will be in form of assignments or workshops.

Objectives

1. To help students learn basic principles of physics.
2. To help students learn basic laws and principles of motions and forces.
3. To study and understand concepts of waves and reflection.
4. To study the concepts of radioactivity, nuclear fission and fusion.

Unit 1
Laws of Motion

Motion and its physical interpretation. Newton's law of motion, Law of conservation of linear momentum and its applications. Equilibrium of concurrent forces. Static and kinetic friction, laws of friction, rolling friction, lubrication. Circular motion: Centripetal and Centrifugal force, examples of circular motion, Projectile motions and its application.

Unit 2
Nature and Properties of Wave Optics

Definition of wave and wave motion. Difference between longitudinal and transverse waves. Nature and properties of electromagnetic waves. Reflection, refraction polarization and diffraction of light. Refractive index and total internal reflection of light. Microscopes and astronomical telescopes (reflecting and refracting) and their magnifying powers. Physical and Chromatic aberrations. Propagation of sound wave and their properties.

Unit 3
Nuclear Physics

Structure of atom. Rutherford's model of atom; Bohr model, energy levels, hydrogen spectrum. Composition and size of nucleus, atomic masses, isotopes, isobars; isotones. Radioactivity – alpha, beta and gamma particles/rays and their properties; radioactive decay law. Mass-energy relation, mass defect; binding energy per nucleon and its variation with mass number; nuclear fission and fusion.

References Books

1. Sears and Zemansky, University Physics, Pearson Publication
2. Paul G Hewitt Conceptual Physics Pearson Publication
3. NCERT Physics Part 1 And 2 NCERT
4. Gomber & Gogia Fundamentals of Physics Pradeep Publications, Jalandhar
5. Arvind Kumar & Shrish Barve, How and Why in Basic Mechanics Universities Press

CTBTCSE SI P2: Engineering Mathematics – 1

Teaching Scheme					Evaluation Scheme								
					Theory						Practical		Total
					Internal Exams				University Exams		University Exams (LPW)		
					TA-1/TA-2		MSE						
Th	Tu	Pr	C	TCH	Marks	Hrs	Marks	Hrs	Marks	Hrs	Marks	Hrs	
3	1	0	4	4	25	00:45	50	01:30	100	03:00	-	-	200

Objectives

1. To learn the concept of higher order derivatives and Partial derivatives
2. To derive series expansion of some standard functions
3. Learn to trace the curves: Cartesian and polar
4. To learn concept of Matrix Algebra and apply it to find solution of system of Linear Equations and eigen value problems and to learn algebra of Complex numbers

Unit 1
Successive Differentiation

Higher order derivatives and some problems based on it. Some standard results on nth derivative and problems based on it. Leibnitz's theorem and problems based on it.

Unit 2
Expansion & Indeterminate Forms

Maclaurin's theorem and problems based on it. Taylor's theorem and problems based on it, Review of limit, L- Hospital's rule, Indeterminate forms $\frac{0}{0}, \frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, $0^0, 1^\infty$

Unit 3
Partial Differentiation & its applications

Limit, Continuity of functions of several variables, Partial derivatives, Total Derivative, Homogeneous Functions, Euler's theorem, Differentiation of implicit functions, Jacobian, error and approximation, maxima and minima

Unit 4
Complex Numbers

Algebra of complex numbers, Modulus, Arguments, Polar Form, De' Moivre's theorem, Expansion of $\cos n\theta$, $\sin n0\theta$, in powers of $\cos\theta$ and $\sin\theta$, Roots of complex

numbers, Solutions of Quadratic equations, Circular functions, Hyperbolic functions, Relation between circular and hyperbolic functions, logarithm of a complex number.

Unit 5
Matrices

Types of Matrices, Elementary row operation, Rank of a matrix, Normal form, Consistency of system of simultaneous linear equations, Inverse of a matrix by Gauss Jordan method, linearly dependent and independent vectors, Eigen values and eigen vectors, Cayley Hamilton theorem.

References Books

1. E. Kreyszig, Advanced Engineering Mathematics (8th Edition), Wiley-India (1999).
2. Higher Engineering Mathematics, Thirty-fifth edition. B. S. Grewal, Khanna Publication.
3. Higher Engineering Mathematics – Vol. 1, Dr. K. R. Kachot, Mahajan Publ. house
4. Thomas' Calculus, Maurice D. Weir, Joel Hass, Frank R. Giordano, Person Education.
5. Calculus, James Stewart, Thomson (5th Edition, 2003).
6. T. M. Apostol, Calculus, Volumes 1 and 2 (2nd Edition), Wiley Eastern (1980).

CTBTCSE SI P3 : Basics of Electrical Engineering

Teaching Scheme					Evaluation Scheme								
					Theory						Practical		
					Internal Exams				University Exams		University Exams (LPW)		Total
					TA-1/TA-2		MSE						
Th	Tu	Pr	C	TCH	Marks	Hrs	Marks	Hrs	Marks	Hrs	Marks	Hrs	
2	1	0	3	3	25	00:45	50	01:30	100	03:00	-	-	200

Objectives

1. To acquaint the students in fundamental concepts of Electrical Engineering. which includes various Engineering Standards, Basic Elements of Circuits and its properties, Basic of Energy Conversion, Various fundamentals and Laws governing the phenomena of Electromagnetism and Electrostatics.
2. To impart training to the students in order to enable them to understand, analyze and evaluate various aspects of basic D.C. as well and A.C. Circuits.
3. To acquaint the students in various fundamentals and Laws governing the phenomena of Electromagnetism and Electrostatics.
4. To lay foundation stone to study the advance concepts of Electrical Engineering and would also be useful in day-to-day application.

Unit 1
Fundamental and Basic Circuit Element

Concepts of E.M.F., Potential Difference and current, Resistance and Resistors, effect of temperature on resistance, resistance temperature coefficient, resistors in series and parallel, S.I. units of work, power and energy.

Unit 2
Electromagnetism

Magnetic effect of an electric current, cross and dot conventions, right 08 hand thumb rule and cork screw rule, nature of magnetic field of long straight conductor, solenoid and toroid. Concept of m.m.f., flux, flux density, reluctance, permeability and field strength, their units and relationships. Simple series and parallel magnetic circuits, comparison of electrical and magnetic circuit, force on current carrying conductors placed in magnetic field, Fleming's left-hand rule. Faradays laws of electromagnetic induction, statically and dynamically induced E.M.F., self and mutual inductance, coefficient of couplings. Energy stored in magnetic field. Charging and discharging of inductor and time constant.

Unit 3

Electrostatics

Electrostatics field, electric flux density, electric field strength, absolute permittivity, relative permittivity, capacitance and capacitor, composite dielectric capacitors, capacitors in series and parallel, energy stored in capacitors, charging and discharging of capacitors and time constant.

Unit 4
D.C. Circuits & AC Fundamentals

Classification of electrical networks, Ohm's law, Kirchhoff's law and their applications for network solutions. Simplifications of networks using series and parallel combinations and star-delta conversions, Sinusoidal voltages and currents, their mathematical and graphical representation, Concept of instantaneous, peak (maximum), average and R.M.S. values, frequency, cycle, period, peak factor and form factor, phase difference, lagging, leading and in phase quantities and phasor representation. Rectangular and polar representation of phasors.

Unit 5
AC Circuits

Single Phase AC Circuit: Study of A.C. circuits consisting of pure resistance, pure inductance, pure capacitance and corresponding voltage-current phasor diagrams and waveforms. Development of concept of reactance, study of series R-L, R-C, R-L-C circuit and resonance, study of parallel R-L, R-C and R-LC circuit, concept of impedance , admittance, conductance and susceptance in case of above combinations and relevant voltage-current phasor diagrams, concept of active, reactive and apparent power and power factor. Single Phase Power Measurement.
Poly-Phase AC Circuit: Poly phase A.C. Circuits: Concept of three-phase supply and phase sequence. Voltages, currents and power relations in three phase balanced star-connected loads and delta-connected loads along with phasor diagrams. Three Phase Power Measurement.

References Books

1. Electrical Technology Vol.1, B.L.Theraja, S.Chand Publication, New Delhi
2. Basic Electrical Engineering, V.N.Mittal, TMH Publication, New Delhi
3. Basic Electrical Engineering, V.K.Mehta, S.Chand and Company Ltd., New Delhi
4. Electrical Technology, Edward Hughes, Seventh Edition, Pearson Education
5. Elements of Electrical Technology, H.Cotton, C.B.S. Publications
6. Basic Circuits Analysis, John Omalley Shawn, Mc Graw Hill.
7. Principles of Electrical Engineering, Del. Toro, PHI

CTBTCSE SI P4 : Fundamentals in Computer Programming with C

Teaching Scheme					Evaluation Scheme								
					Theory						Practical		
					Internal Exams				University Exams		University Exams (LPW)		Total
					TA-1/TA-2		MSE						
Th	Tu	Pr	C	TCH	Marks	Hrs	Marks	Hrs	Marks	Hrs	Marks	Hrs	
3	0	0	3	3	25	45 Mins	50	1.5	100	3	-	-	200

Objectives

1. To provide an entry-level computer programming concept.
2. To develop the Logical and programming ability in students.
3. To improve their proficiency in applying the computing fundamentals to their field of study.
4. To provide a logical base to high level programming languages.

Unit 1
Fundamentals of Computer

What is computer? History of computer, Block diagram of computer system, Hardware and software, Overview of types of operating systems, Compiler and interpreter, Programming Languages, Flowchart and Algorithm.

Unit 2
Overview of C

History of C, Features of C, Basic structure of C program, Process of executing a C program, Character set, trigraph sequences, C tokens, Data type, Variable, Storage class, Symbolic Constant, Overflow of data, Operators, Operator Precedence and Associativity, Type conversions, I/O Functions.

Unit 3
Branching & Looping statements

Introduction, if statement, types of if statement, switch statement, while statement, for statement, do-while statement, go to statement, break and continue statement

Unit 4
Array & Structure

Introduction, One-dimensional array, Two-dimensional array, multidimensional array, limitation of array, strings, string handling functions, table of string, defining a structure, declare and accessing structure variable, structure member as array, structure variable as array, structure within structure, unions, bit fields.

Unit 5
Pointers and User Defined Functions

Introduction, advantages, declaration of pointers, chain of pointers, scale factor, pointers and arrays, pointers and structures, Advantages of functions, elements of functions, categories of functions, recursion, function and arrays, functions and structure, functions and pointers.

References Books

1. Programming in ANSI C, E. Balagurusamy, PHI
2. Let us C, Yaswant Kantikar, BPB
3. C: The Complete Reference, Herbert Schildt, McGrawHill
4. Computer concepts and Programming, Vikas Gupta, DreamTech
5. Computer fundamentals and Programming in C, Pradip dey and Manas Ghosh, Oxford.

CTBTCSE SI P5: Engineering Graphics

Teaching Scheme					Evaluation Scheme								
					Theory						Practical		
					Internal Exams				University Exams		University Exams (LPW)		Total
					TA-1/TA-2		MSE						
Th	Tu	Pr	C	TCH	Marks	Hrs	Marks	Hrs	Marks	Hrs	Marks	Hrs	
2	0	0	2	2	25	45 Mins	50	1.5	100	3	-	-	200

Objectives

1. To allow and help students visualize the objects.
2. To allow and help students understanding and read drawing.
3. To allow and help students to draw and present their drawing.
4. To allow and help students in understanding 3D Projections.

Unit 1
Introduction & Geometric Constructions

Introduction: Drawing instruments & accessories, drawing layout and lettering, drawing conventions and dimensioning rules, geometric constructions, BIS, representative fraction. **Types of graphical scales:** (plain scale, diagonal scale, Vernier scale, comparative scale, scale of chords). **Engineering curves classifications and application:** [parabola, ellipse, hyperbola, cycloidal curves: (spirals, roulettes, trochoids, involutes)

Unit 2
Projection of points & lines

Basics of projection theory: Principals of projection, Projection methods (1st & 3rd angle), Planes of projections, Symbols for method of projection, **Projection of points**: Introduction, Projection in quadrants, **Projection of lines:** Introduction, Inclined to HP and VP, True length & inclinations, Traces of lines

Unit 3
Projection of planes & solids

Projection of planes: Introduction, Types of planes, Projections of a Plane surface & its examples (A. perpendicular to both the reference planes, B: Perpendicular to one plane and parallel to another plane, C: Perpendicular to one plane and inclined to the other plane, D: Inclined to both planes)

Unit 4
Projection & Section of solids

Projection of solids: Introduction, Types of solids, Projection of solids (Cylinder, Cone, Pyramid and Prism), **Sections of Solids:** Introduction, Sections of Solids, A: Section plane parallel to VP, B: Section plane parallel to HP, C: Section plane perpendicular to HP and inclined to VP, D: Section plane perpendicular to VP and inclined to HP, Intersections of solids, Methods of Intersection, Development of Surfaces, Methods of development, True shape of the section

Unit 5

Orthographic & Isometric Projections

Isometric Projection: Introduction, Classifications, Types of projection, 1st angle projection, 3rd angle projection, Projection of points, lines & planes, Isometric axes, Lines & planes, Isometric scale, Isometric projection and Isometric view, Conversion of Isometric to Orthographic Projections

References Books

1. Engineering drawing NSQF, Directorate General of Training, Ministry of Skill Development & Entrepreneurship, Government of India.
2. Dhananjay A Jolhe, Engineering Drawing, Tata McGraw Hill.
3. M.B. Shah and B.C. Rana, Engineering Drawing, Pearson Education.
4. K. Venugopal, Engineering Drawing and Graphics, New Age International (P) Ltd., publishers.
5. N.D. Bhatt and V.M. Panchal, Engineering Drawing, Charotar Publishing house.
6. Engineering Graphics, P.J.Shah, S.Chand & Company Ltd., New Delhi
7. Engineering Drawing, P.S.Gill, S.K.Kataria & sons, Delhi
8. Engineering Drawing, B. Agrawal and C M Agrawal, Tata McGraw Hill, New Delhi

CTBTCSE SI P6: Communication Skills

Teaching Scheme					Evaluation Scheme								
					Theory						Practical		
					Internal Exams				University Exams		University Exams (LPW)		Total
					TA-1/TA-2		MSE						
Th	Tu	Pr	C	TCH	Marks	Hrs	Marks	Hrs	Marks	Hrs	Marks	Hrs	
2	1	0	3	3	25	00:45	50	01:30	100	03:00	-	-	200

Objectives

1. Communicate across the cultures in professional groups and develop their critical listening skills.
2. Improve their competences in professional writing. Use grammar & vocabulary in correct and appropriate manner.
3. Articulate different sounds in effective ways.
4. Improve their speaking skills in day-to-day life.

Unit 1
Basic Fundamentals of Communication

Meaning, definition, objectives & Characteristics /nature of Communication, Communication Process, flow of Communication, Introduction to Professional Communication, Principles of Professional Communication, Communication Networks, Informal Communication

Unit 2
Non-Verbal Communication

Introduction, Components of NVC - Proxemics, Haptics, Kinesics, Chronemics, Paralinguistic Features, Other NVC, Importance of NVC in oral communication, **Listening Skills:** Hearing v/s Listening, Types & Barriers of Listening, Active v/s Passive Listening, Traits of a good listener

Unit 3:
Writing Skills

Paragraph Development: What is Paragraph? Components of Paragraph – Unity, Topic Sentence, Cohesion, Coherence, Adequate Development, Approaches of Paragraph – Inductive , Deductive & Expository Approach, Types of Paragraph, Attributes of good paragraph, Use of Transitional Words, **Business & Technical Letter Writing**: Introduction to Letter Writing, Informal Letters, Personal Letter Vs Business Letter, Style of writing Business Letter, Principles of writing Business Letter, Layout of Business Letter, Types of Letter – Inquiry, order, quotation, claim & adjustment, sales ,credits & Circular

Unit 4
Verbal Communications

Reading Skills: Purpose of reading, Understanding and Interpreting Ideas, Reading Rates, Reading & Interpretation, Interpreting Technical graphics in reading, Techniques of Reading, Reading Comprehensive Skills, Reasons for Poor the Reading Comprehensive Skills, **Conversation Practice:** Dialogue Practice, Telephonic Conversation, Speaking for Various Purposes, **Grammar & Vocabulary:** Parts of Speech, Tenses, Active Voice & Passive Voice, Confusable words

Unit 5
Language through Literature

Selected Stories from "Wise and Otherwise" by Sudha Murthy 1) A Lesson in Life from a Beggar 2) Death without Grief 3) Idealists at Twenty Realists at Forty 4) Think Positive, Be Happy 5) Crisis of Confidence 6) Sorry, the line is Busy 7) Oh Teacher, I Salute Thee 8) Life is an Examination

References Books

1. Green Andy, Effective Personal Communication Skills for Public Relations, Kogan Page, Limited, 2006.
2. John M. Penrose, Jr., Robert W. Rasberry, Robert J. Myers, Advanced Business Communication, Thomson/South-Western, 2004
3. Technical Communication, D.K.Chakradev, Tech-max publication.
4. Basic Business Communication, Flatly and Lesicar
5. Basic Communication Skills for Technology, Andrea J. Rutherford, Pearson Education
6. From sentence to paragraph, William J. Kelly and Deborah L. Lawton, Longman
7. Technical Communication: Principles and Practice, Meenaxi Raman and Sangeeta Sharma, Oxford Press
8. Wise and Otherwise, SudhaMurty, Penguin Books India Pvt. Ltd. Delhi
9. An Intermediate English Grammar, Raymond Murphy, Cambridge University Press
10. A High School English Grammar, Wren & Martin, S. Chand Publication